

CIS530 HW4: Submission Example

5. Free-response Questions (10 points)

Results

- **Task:** Clustering Shakespeare's Plays using Vector Representations
- **Results:** I computed similarity scores between Shakespeare's plays using three different vector representations:

1. Raw Term-Document matrix
2. TF-IDF weighted Term-Document matrix
3. PPMI-weighted Term-Context matrix (aggregated by play)

Each representation was evaluated with three similarity functions: Cosine, Jaccard, and Dice. Cosine similarity produced the most consistent and interpretable results across all representations. Using this setup, I generated a similarity matrix for all 36 plays and applied hierarchical clustering.

With TF-IDF + Cosine similarity, clusters showed strong alignment with traditional Shakespearean categories:

Cluster 1 (Comedies): "As You Like It", "Twelfth Night", "Much Ado About Nothing", etc.

Cluster 2 (Tragedies): "Hamlet", "Othello", "King Lear", "Macbeth"

Cluster 3 (Histories): "Henry IV Part 1", "Henry V", "Richard III", etc.

In contrast, the raw term-document vectors often grouped unrelated plays due to high-frequency stop words, while PPMI vectors showed strong thematic similarity but less clear genre separation.

Discussion:

This clustering experiment shows that vector representations, especially TF-IDF weighted with cosine similarity, can capture high-level semantic distinctions among Shakespeare's plays. The alignment of clusters with known genres (comedies, tragedies, histories) indicates that the word usage patterns specific to each genre are reflected in the underlying word distributions.

Interestingly, PPMI-based clustering revealed more nuanced thematic connections — for example, it grouped "Titus Andronicus" with "Macbeth" due to shared vocabulary related to violence and revenge, even though they are traditionally separate in genre. This suggests that PPMI captures deep semantic usage beyond genre boundaries.

Overall, TF-IDF + Cosine is best for recovering known categorical structure, while PPMI offers insight into shared themes or motifs across plays.

Extra Credit (15 points)

Results

- **Task:** Comparing similarity metrics and vector representations in word ranking tasks.
- **Results:** I evaluated how well different similarity metrics (cosine, Jaccard, Dice) performed when applied to word vectors derived from various weighting schemes (PPMI and TF-IDF) built from a term-context matrix. Using the SimLex-999 dataset as a reference, I computed Kendall's Tau correlation to assess alignment with human judgments. For example:
 - PPMI + Cosine: Kendall's Tau = 0.47
 - PPMI + Jaccard: Kendall's Tau = 0.29
 - TF-IDF + Cosine: Kendall's Tau = 0.35
- **Discussion:**

From the experiments, I also found that cosine similarity generally produced the highest alignment with human similarity scores across both PPMI and TF-IDF representations. PPMI outperformed TF-IDF overall, likely because it emphasizes informative co-occurrences by removing the influence of highly frequent context terms. Jaccard and Dice metrics, while useful for binary or sparse data, appeared to be less effective for dense vector comparisons.

This also suggests that PPMI + Cosine is a strong default choice for tasks involving semantic similarity. Additionally, we observed that concrete nouns (like "car" and "automobile") were more consistently ranked well compared to abstract terms, highlighting the limits of distributional semantics for more subjective or context-sensitive concepts.